DEEP-DIVE ARCHITECTURAL & LOGIC AUDIT #03F: MARTIAL ARTS & TECHNIQUES (PART 6 — SUPPORT ARTS, HEALING, BREAKTHROUGH PROTECTION, DISTRESS SIGNALS & PROFESSION/LIFESTYLE AUXILIARY ARTS)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` / `SIM.PATCH_VERSION = '2.6.0'` / `HC.FINGERPRINT = '155:2f9c9dd7'`)
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini** Models)
- **Series Context**:
  - `03A`: Active Combat Moves (`COMBAT_TECHNIQUE`), `SK.list` Caching, `chooseSkill` AI, 16 Weapons & Spatial Range.
  - `03B`: Composable Engine, 34 Elemental Reactions (`DARK_YIN` vs `YIN_SHADOW`), & Duet Synergies.
  - `03C`: Mantras & Inner Qi Circulation (`MANTRA` / `MANTRA_SYSTEM`), Lifespan Traps & Sect Formation Warfare.
  - `03D`: Qinggong, Sword Flight, Celestial Levitation, Stealth/Recon Arts & Combat Mobility.
  - `03E`: Defensive Kungfu (`DEFENSIVE_ARTS`), Passive Slots (`passiveLoadout`), Innate Arts & Passive Constellations.
  - **`03F` (This Document)**: Dedicated, exhaustive deep dive into **Support & Healing Arts (`SUPPORT_TECHNIQUE` / `CULT.SUPPORT_ARTS`), Training & Rescue Aids (`role: 'HEAL_SUPPORT_TRAINING'`), and Profession/Lifestyle Auxiliary Scriptures (`AUXILIARY_ART` / `role: 'PROFESSION_LIFESTYLE'` / `AUX_*` — 160+ total manuals)**.

---

0. Hard Architectural Invariants for AI Implementers

1. **Immutable `human_core.js`**: Never modify `GAME_SOURCE/js/human_core.js` (`HC.FINGERPRINT === '155:2f9c9dd7'`).
2. **40-Script Load Order & Closed-Loop Economy**: Preserve exact `<script>` order in `index.html`. Profession bonuses that save materials or boost yield must respect closed-loop resource caps (`W.world.resourcePool`) and never mint free Spirit Stones out of thin air.
3. **100% Emergent NPC Autonomy**: Physicians (`healer`), Alchemists (`alchemist`), Smiths (`smith`), Herbalists, Scholars (`scholar`), Musicians, and Strategists must

autonomously study, equip, and switch Support/Auxiliary postures and trigger emergency sect relief without player UI clicks.

4. **Test-Lock Awareness**: `test/v570_support_and_auxiliary_arts_suite.js:66-69` asserts `44 SUPPORT_TECHNIQUE + 53 AUXILIARY_ART = 97` in `JD.TECH.techniques`. Keep `t.cat` counts intact while adding structured runtime descriptors!

---

1. [SUP-PROF-01] 64+ Profession & Lifestyle Manuals (PROFESSION_LIFESTYLE & AUX_*) Have 0% Runtime Wiring!

1.1 Repository-Wide Verification: Complete Disconnect From economy.js, explore.js, wildlife.js & expedition.js

- **Locations**: `data/data_tech.js` (`ART190_XIAN_065..072`, `ART190_BODY_065..072`, `ART190_GOD_065..072`, `AUX_PILL_NINE_REVOLUTIONS`, `AUX_WEAPON_HEAVEN_FORGE`, `AUX_ARRAY_STARS`, `AUX_BEAST_BLOOD_PACT`), `data/data_martial.js` (`v2.0.34 AUXILIARY` manuals), `js/skills.js:183`, `js/economy.js`
- **Root Cause**:
  1. `SK.profile(t)` (`js/skills.js:183`) explicitly returns **`null` (`// วิชาอาชีพ/ไม่ใช่การรบ`)** for every `role === 'PROFESSION_LIFESTYLE'` and `AUX_*` manual.
  2. None of those 28 core manuals in `data_tech.js` define a `t.passive` object, so `MARTIAL.passiveStats(n)` ignores them completely.
  3. Searching the entire repository for `'PROFESSION_LIFESTYLE'`, `'AUX_PILL_NINE_REVOLUTIONS'`, `'AUX_WEAPON_HEAVEN_FORGE'`, `'AUX_ARRAY_STARS'`, `'AUX_BEAST_BLOOD_PACT'`, and even `'SUPPORT_ARTS'` inside `js/economy.js`, `js/wild_resources.js`, `js/wildlife.js`, and `js/external_market.js` returns **ZERO matches (`0` hooks!)**!
- **Every Broken Profession Manual Group & What Players/NPCs Were Promised**:

Group	Exact Technique IDs in `data_tech.js`	Explicit Thai Lore Promise (`moves` / `nameTh`)	Actual Effect in `JS2-v2.6.0`
Alchemy & Furnace Control	`ART190_XIAN_065`, `ART190_BODY_066`,	Stabilize furnace fire, increase pill quality (`q`) & rare pill success	**`0%` effect** on `ECO.successChance`,

Group	Exact Technique IDs in `data_tech.js`	Explicit Thai Lore Promise (`moves` / `nameTh`)	Actual Effect in `JS2-v2.6.0`
	`AUX_PILL_NINE_REVOLUTIONS` (T5)	chance, reduce fuel/herb waste	`ECO.finishCraft`, or pill `q`
Smithing & Talisman Inscription	`ART190_BODY_065`, `ART190_GOD_066`, `AUX_WEAPON_HEAVEN_FORGE` (T5), `ART190_XIAN_066`, `ART190_GOD_065`	Boost forged weapon/armor `q` & `maxDur`, reduce forge failure, save cinnabar/talisman paper	**`0%` effect** on `A.forge`, `ECO.finishCraft`, or talisman output
Soil Detox, Pollination & Farm Wards	`ART190_XIAN_068`, `069`, `072`; `ART190_BODY_070`, `072`; `ART190_GOD_067`, `068`, `072`	Purify poisoned soil, boost rare herb yield & quality, trap/repel wild beasts raiding herb gardens	**`0%` effect** on `A.farm`, `ECO.harvestHerb`, or `WILD` garden raids
Cargo Preservation & Spirit-Vessels	`ART190_XIAN_070`, `071`; `ART190_BODY_071`; `ART190_GOD_070`, `071`	Ice/divine seal prevents herb/organ spoilage on trade routes; reduce supply/stone cost of caravans & airships	**`0%` effect** on `dispatchSupplyConvoy`, `EXT` trade value, or `EXPED` supply burn
Beast Taming, Aura & Recovery	`AUX_BEAST_BLOOD_PACT` (T5), `ART190_BODY_067`, `068`; `ART190_GOD_069`	Calm wild beasts while gathering, boost `attemptTameBeast` odds, accelerate spirit-pet `bond` & `fatigue` recovery	**`0%` effect** on `CULT.attemptTameBeast` or pet `fatigue`/`bond`
Array & Leyline Engineering	`AUX_ARRAY_STARS` (T6), `ART190_XIAN_067`, `ART190_BODY_069`	Repair sect barrier cracks, boost protective array core efficiency & mining vein yield	**`0%` effect** on `SECT.ELEMENTAL_BARRIER`, `LEYLINE_GRID`, or `A.mine`

1.2 Remediation Blueprint ([SUP-PROF-01])

Build a single cached O(1) authority **`CULT.PROFESSION_ARTS.getModifiers(n)`** (invalidated on `n._passVer` / `n.cult.techs.length`) that aggregates learned `PROFESSION_LIFESTYLE` and `AUX_*` manuals (scaled by `tier` and `c.tm[id]`):

- **`economy.js`**: Add `alchemySuccessBonus` (`+4%..+12%`), `pillQualityBonus` (`+0.04..+0.12`), `forgeSuccessBonus` (`+4%..+12%`), `gearMaxDurBonus` (`+6..+18`), and `herbYieldChance` (`+8%..+22%` bounded by `resourcePool`).
- **`wildlife.js` & `ai.js` (`A.farm` / `A.mine`)**: Farm-ward manuals (`XIAN_072`, `BODY_072`, `GOD_072`) learned by farmers/guards place active perimeter wards on herb plots that repel/damage encroaching wild beasts; mining/earth-vein manuals (`BODY_069`, `XIAN_067`, `AUX_ARRAY_STARS`) boost `A.mine` yield and passively regenerate `SECT.ELEMENTAL_BARRIER.barrierHp` when the `arraymaster` works!
- **`expedition.js` & `external_market.js`**: Cargo-preservation & hauling manuals (`XIAN_070/071`, `BODY_071`, `GOD_070/071`) reduce expedition supply consumption by `-12%..25%` and grant `+8%..15%` freshness value on exported herbs/beast organs!
- **`cultivation.js` (Pets)**: Beast-pact manuals (`AUX_BEAST_BLOOD_PACT`, `BODY_067/068`, `GOD_069`) add `+10%..18%` to `CULT.attemptTameBeast` and `+35%` hourly pet `fatigue` recovery!

---

2. [SUP-TRAIN-02] 36 HEAL_SUPPORT_TRAINING Manuals Flattened Into Generic Combat Heals (Missing 4 Iconic Xianxia Loops!)

2.1 Distinct Systemic Roles Erased by `skills.js:173`

In `data/data_tech.js`, entries `053..064` across `XIAN`, `BODY`, and `GOD` (`36` manuals) were designed around **five distinct sect-life pillars**. Currently, 12 of them (`053, 056, 059, 062` tagged `*_MANTRA`) return `SK.profile === null`, and the remaining 24 (`AUXILIARY_ART`) all become identical `arch: 'heal'` combat heals! Four iconic emergent Xianxia interactions described verbatim in their `moves` are completely missing:

1. **Joint Heavenly Tribulation Protection (護法 / คุ้มครองการทะลวงด่าน — `ART190_XIAN_056`, `ART190_BODY_056`, `ART190_GOD_057`)**:
   - **Lore**: *"กางม่านปราณช่วยศิษย์ แบ่งรับสายฟ้าทัณฑ์ คุ้มครองการทะลวงขั้น"*.
   - **Emergent Design**: When an NPC enters `A.breakthrough` (`stage >= 3`), if their Mentor, Taoist Companion, or benevolent Elder (`empathy >= 60` / `loyalty >= 65`) knows a Tribulation-Protection Art and is idle/nearby, they

autonomously perform **Dharma-Protector Duty (`guardBreakthrough`)**—absorbing `25%` of tribulation lightning backlash (taking minor HP/Qi drain themselves) to grant the breakthrough candidate **`+8%..+14%` `breakthroughChance`** and `-50%` failure injury severity! (And bind a `GOLD_THREAD` if they save a critical breakthrough!)

2. **Enlightenment Teaching & Spar-Tempering (**ชี้แนะแก่นแท้เบิกปัญญา **— `ART190_XIAN_057`, `ART190_BODY_057`, `ART190_GOD_058`)**:
   - **Lore**: *"ชี้แนะแก่นแท้กระบวนท่า คลายปมสงสัยในเต๋า เร่งความเข้าใจวิชา"*.
   - **Emergent Design**: Wire into `AI.pairTick` (`type === 'teach'`) and `C.sparEnd`: if the senior partner knows an Enlightenment-Teaching Art, boost the junior's `skillPractice` / `c.tm` gain by **`+30%`** and grant a small `bn` bottleneck insight boost!
3. **Emergency Distress Sword-Flare / Earth-Drum / Beacon (**ส่งสัญญาณขอความช่วยเหลือ **— `ART190_XIAN_064`, `ART190_BODY_064`, `ART190_GOD_064`)**:
   - **Lore**: *"ยิงปราณกระบี่ขึ้นฟ้า แจ้งพิกัดแก่ศิษย์ร่วมสำนัก เรียกกำลังเสริมทันท่วงที"*.
   - **Emergent Design**: When an outnumbered/low-HP (`hp < 50%`) sect member is fighting a `wild` beast or `raid` vanguard on the Sect Map (`f.kind === 'wild' || f.kind === 'raid'`) and knows a Distress-Signal Art (once per fight), fire a signal flare (`fx: '🎆 สัญญาณกระบี่ขอความช่วยเหลือ!'`) that pulls up to `2` healthy patrolling/idle sectmates within `45` tiles into the fight via `COMBAT.join(W, f, ally.id, 'A')`!
4. **Spare-Qi Jade / Marrow / Prayer-Pearl Reserve (**เก็บกักปราณสำรองยามว่าง **— `ART190_XIAN_061`, `ART190_BODY_061`, `ART190_GOD_059`)**:
   - **Lore**: *"บรรจุปราณยามว่าง ดึงปราณสำรองยามคับขัน"*.
   - **Emergent Design**: During peaceful meditation at `100%` Qi, bank surplus Qi into `n.ext.qiReserve` (up to `30%` `qiMax`); when `qi < 25%` in combat, automatically burst-restore `n.ext.qiReserve` into `n.h.state.qi`!

---

3. `[SUP-ERA5-03]` Five Live Runtime Bugs in `CULT.SUPPORT_ARTS` (`v5.7.0` in `combat.js`, `economy.js`, `ai.js`)

3.1 Bug 3A — 100% Dormant Without Manual UI Clicks (`setSupportPosture` & `triggerSectEmergencySupport`)

- **Verified Fact**: Repository-wide search confirms `setSupportPosture` and `triggerSectEmergencySupport` are called **exclusively in `cultivation.js`, `ui.js`, and `v570` test**.

- **Impact**: In natural play, `n.supportState` is `undefined` on 100% of NPCs! Neither `stepFacilityAndDailySynergy` nor Battlefield Support Auras ever run, and the Sect's emergency Heart-Demon/Plague relief button (`trigger-sect-emergency-support` in `UI.rSect`) never triggers unless clicked by the human player!

3.2 Bug 3B — Three Unwired Spec Promises in `PATCH_IMPLEMENTATION_REPORT_V5_7_0.md`

1. **`ALCHEMICAL_INCENSE_MIST` (`+22%` Alchemy/Forging/Kitchen crafting efficiency)**: `economy.js` contains **zero** references to `SUPPORT_ARTS` or `ALCHEMICAL_INCENSE_MIST`! Wire nearby/self `ALCHEMICAL_INCENSE_MIST` aura into `ECO.successChance` (`+0.08`), craft speed (`+18%`), and `A.cook` output!
2. **`DIVINATION_STAR_GUIDANCE` (`+22%` Scripture Pavilion study speed)**: Wire into `A.study.tick` (`js/ai.js`) so scholars/elders with `DIVINATION_STAR_GUIDANCE` accelerate `learning.got` for themselves and nearby readers in `scripture`!
3. **`QIMEN_BATTLE_BANNER` (`+30%` Sect Guard security)**: Wire into `A.guard.tick` (`js/ai.js`)!

3.3 Bug 3C — Pre-Damage Overheal Waste & `healBlock` Bypass in `COMBAT.applyDamage` (`combat.js:356–388`)!

- **Exact Code (`combat.js:368–386`)**:

```javascript
const healAmt = Math.max(1, Math.round(C.hpMax(d) * sup.effHealPct));
C.setHp(d, Math.min(C.hpMax(d), C.hp(d) + healAmt)); // <--- LINE 382 (BEFORE dmg IS SUBTRACTED AT LINE 437!)
```

- **Four Concrete Bugs Here**:
  1. **Wasted Heal at Full HP (`Order-of-Operations Bug`)**: Line 382 heals `d` **before** incoming `dmg` is subtracted at line 437! When `d` is at `100%` HP (`1000/1000`) and takes a `250` damage hit, `Math.min(1000, 1000 + 80)` clamps to `1000` (**healing `0` HP!**), yet `supNpc` still loses `-2 Qi`, triggers the 2-tick ICD, logs fake `totalAlliesHealed += 80`, and then line 437 drops `d` to `750`! Only trigger the reactive heal when `C.hp(d) - dmg < C.hpMax(d) * 0.92` (applying after or offsetting `dmg`)!
  2. **Bypasses `C.healMultiplier` (`healBlock` / `stygianFreeze` / `ashenToxin`)**: Line 381 never multiplies `healAmt` by `C.healMultiplier(d, nowT)`! **Same bug exists on `arch === 'command'` (`combat.js:929`) and `arch === 'harmonize'` (`combat.js:944`)**—all three heal 100% right through `stygianFreeze` (`0%` heal) and `healBlock` (`25%` heal)!

3. **Stunned / Silenced Support Casters Project Active Auras**: Lines 359 & 369 do not check `ST.has(supNpc, 'stun' | 'paralyze' | 'silence' | 'cryoStasis', nowT)`. An unconscious or silenced musician keeps playing zither buffs and throwing acupuncture heals!
4. **DoT-Tick Spam & `O(N_npcs)` Scan**: Restrict aura scan to `C.alliesOf(W, f, d.id)` and skip passive DoT ticks (`isDotTick`).

3.4 Bug 3D — Transient `_btBonusPct` Lost on Save/Load or Never Expiring (`v570` line 108)

- `stepFacilityAndDailySynergy` stores Breakthrough Stability Blessing on `elder._btBonusPct` (an underscore-prefixed field without an expiry timestamp `_btBonusUntil`). Store it cleanly as `n.ext.supportBlessing = { btBonusPct, until: W.t + 1440 }` so it persists across `SIM.serialize` / `SIM.deserialize` and expires cleanly after 24 in-game hours!

---

4. `[SUP-RANGE-04]` Forgotten `maxRange` Argument on `lowestAlly(W, f, a)` (`combat.js:535, 649, 897`)!

4.1 Single-Target Heals Still Reach 85+ Tiles Across the Map Despite `TECH-P1-10`!

- **Location**: `js/combat.js:718` (`lowestAlly`) vs `lines 535, 649, 897`
- **Exact Code**:

```javascript
// Line 718: maxRange parameter was added...
function lowestAlly(W, f, a, maxRange) {
  const lim = Number.isFinite(maxRange) && maxRange > 0 ? maxRange : Infinity;
  ...
}
// ...YET Line 895-897 forgot to pass maxHealR into lowestAlly!!
const maxHealR = p.globalRange ? 999 : Math.max(6.5, (hSpec.max || 4.5) * 1.8);
const healTargets = p.aoe ? (inRangeAllies.length ? inRangeAllies : [a])
  : [(() => { const ally = lowestAlly(W, f, a); return ...; })()];
```

- **Consequence**: `lowestAlly(W, f, a)` receives `maxRange = undefined` $\to$ `lim = Infinity`! Single-target `heal` (`line 897`), `command` (`line 924`), and `harmonize` (`line 938`) still heal/buff allies **85+ tiles away across the map**, completely bypassing `maxHealR`!

- **Companion Movement Fix Needed in `fightStep` (`combat.js:600`)**:
    - Once `lowestAlly(W, f, a, maxHealR)` properly enforces `maxHealR`, if a `HEALER` has a ready heal and a wounded ally (`hpFrac < 0.72`) exists in `C.alliesOf(W, f, a.id)` *outside* `maxHealR`, `fightStep` must **walk the healer toward that wounded ally** (`moveToward(W, n, woundedAlly.x, woundedAlly.y)`) rather than walking into the enemy frontline!

---

5. `[SUP-AUTO-05]` Emergent NPC Autonomy Blueprint for Support, Healing & Profession Arts

2. **Specialist Study Priority in `COG.techPreference(n, t)` (`js/cognition.js`)**:
    - Give targeted `+16..+24` study priority on matching `PROFESSION_LIFESTYLE` / `AUX_*` / `SUPPORT_ARTS` manuals to NPCs holding sect offices or high career affinities:
        - `alchemist` / high `ALCHEMY` $\to$ Alchemy fire & incense manuals (`XIAN_065`, `BODY_066`, `AUX_PILL_NINE_REVOLUTIONS`, `SUP_ALCHEMICAL_SPIRIT_MIST_T2`)
        - `smith` / high `FORGING` $\to$ Heaven-Forge & Eternal-Flame manuals (`BODY_065`, `GOD_066`, `AUX_WEAPON_HEAVEN_FORGE`)
        - `healer` / `SAGE_SUPPORT` $\to$ Acupoint, Jade-Spring & Bodhi manuals (`SUP_GOLDEN_NEEDLE_MERIDIAN_T3`, `XIAN_053/054/058`, `GOD_053/054`)
        - `arraymaster` / `general` $\to$ Qimen Banner & Star-Array manuals (`SUP_QIMEN_COMMAND_BANNER_T4`, `AUX_ARRAY_STARS`, `XIAN_067`)
        - `beast_warden` / `BEAST_SOVEREIGN` $\to$ Beast-Pact manuals (`AUX_BEAST_BLOOD_PACT`, `BODY_067/068`, `GOD_069`)
        - `quartermaster` / `envoy` $\to$ Cold-Storage, Hauling & Spirit-Vessel manuals (`XIAN_070/071`, `BODY_071`, `GOD_070/071`)
3. **Autonomous Support Posture Switching (`CULT.SUPPORT_ARTS.evaluateAutonomousPosture(W, n, f)`)**:
    - Auto-initialize `n.supportState` when learning any `SUPPORT_TECHNIQUE` / `AUXILIARY_ART` (and update `W._hasActiveDefSupportAura`).
    - **In Sect Daily Life**:
        - Switch to `'BENEVOLENT_HEALER_AURA'` when working at `infirmary` or near injured/poisoned sectmates;
        - Switch to `'SOUL_MELODY_HARMONY'` at `tea` / `harmonic_pavilion` / `main_hall` when nearby sectmates have `heartDemon >= 25` or `stress >= 55`;

        - Run `stepFacilityAndDailySynergy(W, n)` automatically every `15` ticks during `A.heal`, `A.leisure` (tea/music), `A.brew`, and `A.cultivate`!
    - **In Combat / Expeditions**: Switch to `'BENEVOLENT_HEALER_AURA'` when any ally drops `< 65%` HP, otherwise `'TACTICAL_ARRAY_COMMAND'`.
4. **Autonomous Officer Emergency Relief (`healer` / `master`)**:
    - During daily officer jurisdiction in `sect.js`, if ≥ 3 sect members have `heartDemon >= 48` or severe injuries and cooldown is ready, let the `healer` or `master` autonomously invoke `CULT.SUPPORT_ARTS.triggerSectEmergencySupport(W, mode)` (spending real Qi/herbs from stock rather than acting as a free UI god-button)!

---

6. Implementation & Verification Checklist for GPT / Gemini (`03F`)

- [ ] **Wire All 64+ `PROFESSION_LIFESTYLE` & `AUX_*` Manuals**: Implement `CULT.PROFESSION_ARTS.getModifiers(n)` and connect to `economy.js` (`brew`, `forge`, `harvestHerb`, `ALCHEMICAL_INCENSE_MIST`), `wildlife.js` (herb-garden trap wards), `expedition.js` / `external_market.js` (cargo preservation & supply reduction), `cultivation.js` (pet taming & fatigue recovery), and `sect.js` (`ELEMENTAL_BARRIER` maintenance).
- [ ] **Implement the 4 Unique `HEAL_SUPPORT_TRAINING` Loops**: Dharma-Protector Tribulation Shielding (`guardBreakthrough`), Enlightenment Teaching (`+30%` `pair` teach/spar mastery), Distress Signal Flare (`COMBAT.join` nearby allies), and Spare-Qi Reserve (`qiReserve`).
- [ ] **Fix `CULT.SUPPORT_ARTS` & Combat Heals**: Pass `maxHealR` into `lowestAlly(W, f, a, maxHealR)` + `command`/`harmonize` range; walk healers toward out-of-range wounded allies; apply `C.healMultiplier` to `SUPPORT_ARTS`, `command`, and `harmonize`; fix pre-damage overheal order in `combat.js:382`; block stunned/silenced aura casters; persist `supportBlessing` with `until` across Save/Load.
- [ ] **Full NPC Autonomy**: Add specialist weights in `COG.techPreference`, auto-initialize & switch `n.supportState.posture`, and wire `healer`/`master` AI to `triggerSectEmergencySupport`.